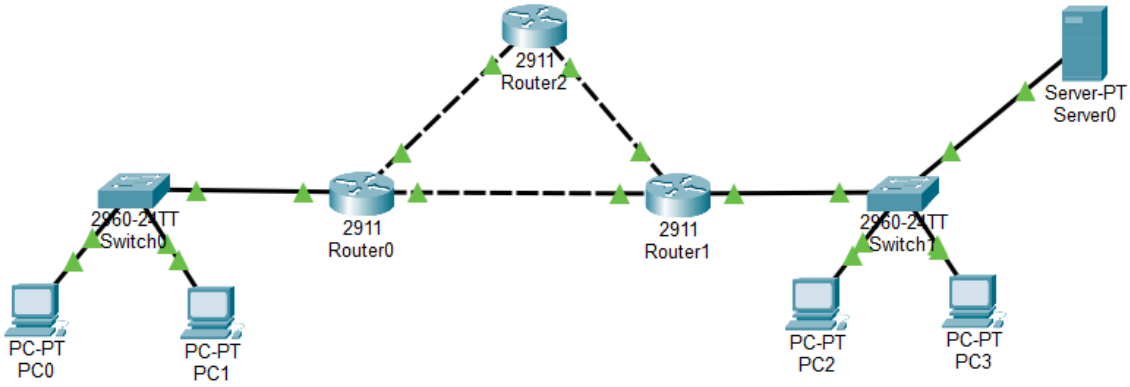
11) a) State routing configuration



Via Router 1

```
C:\>tracert 30.0.0.10

Tracing route to 30.0.0.10 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.0.0.1
  2    0 ms      0 ms      1 ms      20.0.0.2
  3    0 ms      0 ms      0 ms      30.0.0.10

Trace complete.
```

Via Router 2

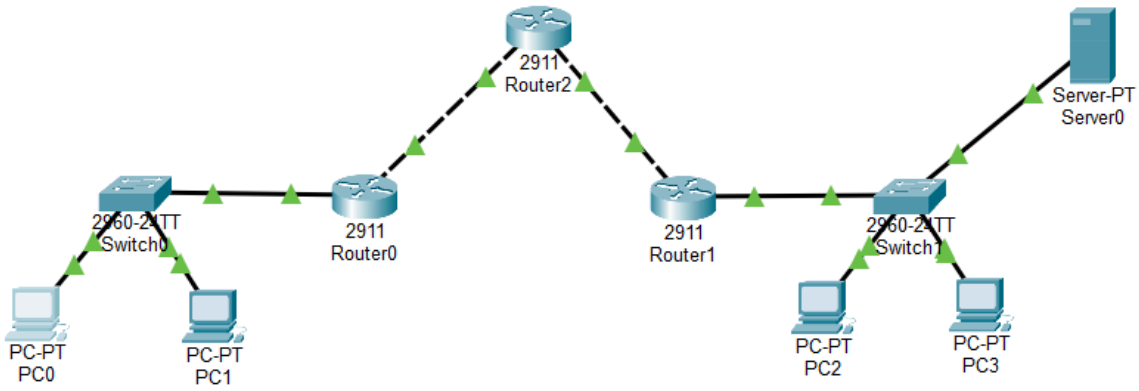
```
C:\>tracert 30.0.0.100

Tracing route to 30.0.0.100 over a maximum of 30 hops:

  1    0 ms      0 ms      5 ms      10.0.0.1
  2    *         0 ms      0 ms      40.0.0.2
  3    *         0 ms      0 ms      20.0.0.2
  4    *         0 ms      0 ms      30.0.0.100

Trace complete.
```

Backup Router



Via Router 1

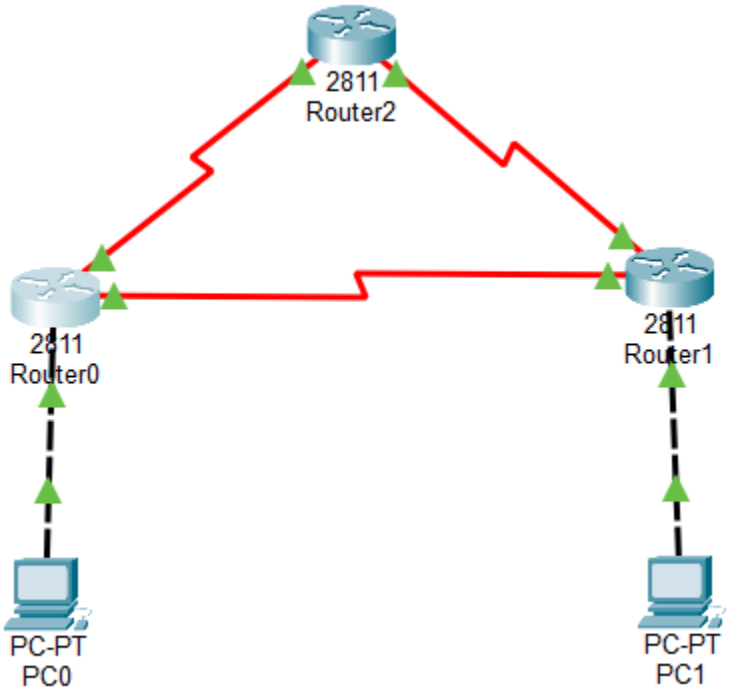
```
C:\>tracert 30.0.0.10

Tracing route to 30.0.0.10 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.0.0.1
  2    0 ms      0 ms      0 ms      40.0.0.2
  3    0 ms      0 ms      0 ms      50.0.0.2
  4    0 ms      0 ms      0 ms      30.0.0.10

Trace complete.
```

11) b) simulate RIP using Cisco Packet Tracer



Route 0:

```
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, FastEthernet0/0
L       10.0.0.1/32 is directly connected, FastEthernet0/0
R    20.0.0.0/8 [120/1] via 192.168.1.250, 00:00:27, Serial0/3/0
     192.168.1.0/24 is variably subnetted, 5 subnets, 2 masks
R       192.168.1.244/30 [120/1] via 192.168.1.250, 00:00:27, Serial0/3/0
                         [120/1] via 192.168.1.253, 00:00:04, Serial0/3/1
C       192.168.1.248/30 is directly connected, Serial0/3/0
L       192.168.1.249/32 is directly connected, Serial0/3/0
C       192.168.1.252/30 is directly connected, Serial0/3/1
L       192.168.1.254/32 is directly connected, Serial0/3/1
```

Route 1:

```
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

R    10.0.0.0/8 [120/1] via 192.168.1.249, 00:00:11, Serial0/3/0
     20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, FastEthernet0/0
L       20.0.0.1/32 is directly connected, FastEthernet0/0
     192.168.1.0/24 is variably subnetted, 5 subnets, 2 masks
C       192.168.1.244/30 is directly connected, Serial0/3/1
L       192.168.1.246/32 is directly connected, Serial0/3/1
C       192.168.1.248/30 is directly connected, Serial0/3/0
L       192.168.1.250/32 is directly connected, Serial0/3/0
R       192.168.1.252/30 [120/1] via 192.168.1.249, 00:00:11, Serial0/3/0
                         [120/1] via 192.168.1.245, 00:00:06, Serial0/3/1
```

Route 2:

```
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

R    10.0.0.0/8 [120/1] via 192.168.1.254, 00:00:27, Serial0/3/1
R    20.0.0.0/8 [120/1] via 192.168.1.246, 00:00:24, Serial0/3/0
     192.168.1.0/24 is variably subnetted, 5 subnets, 2 masks
C       192.168.1.244/30 is directly connected, Serial0/3/0
L       192.168.1.245/32 is directly connected, Serial0/3/0
R       192.168.1.248/30 [120/1] via 192.168.1.246, 00:00:24, Serial0/3/0
                         [120/1] via 192.168.1.254, 00:00:27, Serial0/3/1
C       192.168.1.252/30 is directly connected, Serial0/3/1
L       192.168.1.253/32 is directly connected, Serial0/3/1
```

PC0:

```
C:\>ping 20.0.0.2

Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time=20ms TTL=126
Reply from 20.0.0.2: bytes=32 time=25ms TTL=126
Reply from 20.0.0.2: bytes=32 time=19ms TTL=126
Reply from 20.0.0.2: bytes=32 time=19ms TTL=126

Ping statistics for 20.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 19ms, Maximum = 25ms, Average = 20ms
```

```
C:\>tracert 20.0.0.2

Tracing route to 20.0.0.2 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.0.0.1
  2    7 ms      8 ms      8 ms      192.168.1.250
  3    7 ms      6 ms      8 ms      20.0.0.2

Trace complete.
```

PC1:

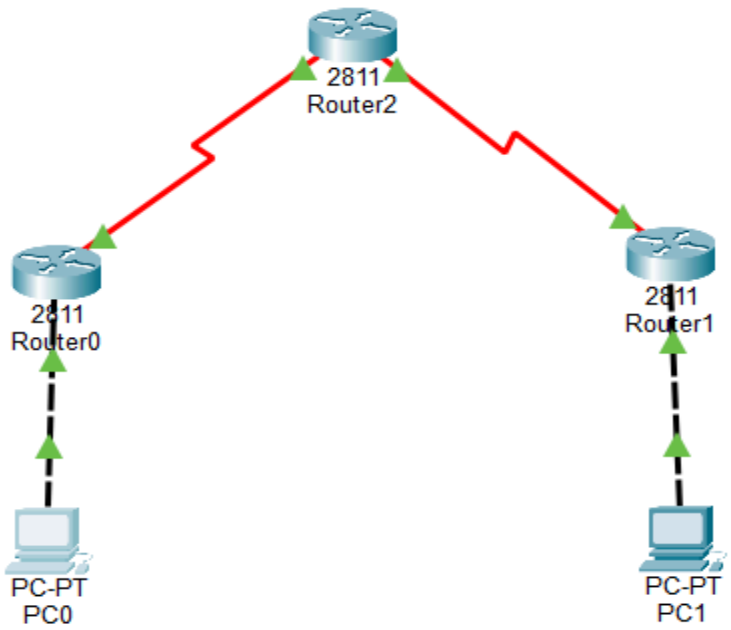
```
C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time=32ms TTL=126
Reply from 10.0.0.2: bytes=32 time=34ms TTL=126
Reply from 10.0.0.2: bytes=32 time=28ms TTL=126
Reply from 10.0.0.2: bytes=32 time=30ms TTL=126

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 28ms, Maximum = 34ms, Average = 31ms
```

Failover:



PC0:

```
C:\>tracert 20.0.0.2

Tracing route to 20.0.0.2 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.0.0.1
  2    9 ms      8 ms      9 ms      192.168.1.253
  3    17 ms     15 ms     16 ms     192.168.1.246
  4    11 ms     13 ms     14 ms     20.0.0.2

Trace complete.
```

Route0:

```
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, FastEthernet0/0
L       10.0.0.1/32 is directly connected, FastEthernet0/0
R    20.0.0.0/8 [120/2] via 192.168.1.253, 00:00:03, Serial0/3/1
     192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
R       192.168.1.244/30 [120/1] via 192.168.1.253, 00:00:03, Serial0/3/1
C       192.168.1.252/30 is directly connected, Serial0/3/1
L       192.168.1.254/32 is directly connected, Serial0/3/1
```